EXAM CRASH-COURSE MASTER GUIDE • ALL PPT CONTENT INCLUDED

Artificial Intelligence Master Study Summary

Complete Technical & Theoretical Revision Notes (20AIC201A / P.Padma Priya Dharishini & Dr. Subarna Chatterjee)

★ CORE DEFINITION & ETYMOLOGY

Artificial Intelligence (AI): Term coined by **John McCarthy in 1956** at the Dartmouth Conference. AI is defined as the design of intelligence in artificial devices—creating machines capable of perceiving, reasoning, learning, adapting, and acting autonomously in dynamic environments.

1. THE FOUR VIEWS OF ARTIFICIAL INTELLIGENCE

AI definitions map along two key dimensions: **Thought processes vs. Behavior** and **Human-like standard vs. Ideal Rationality**.

DIMENSION	HUMAN-LIKE STANDARD	RATIONALITY STANDARD (IDEAL)
Thinking	**Thinking Humanly:** Cognitive modeling approach. Emulates human mental faculties and thought traces via computational models, psychology experiments, and brain imaging (e.g., General Problem Solver - GPS by Newell & Simon, 1961).	**Thinking Rationally:** "Laws of Thought" / Logicist approach. Codifies irrefutable reasoning using formal logic and syllogisms (Aristotle: *Socrates is a man; all men are mortal → Socrates is mortal*). Uses logic notation to yield correct conclusions from true premises.
Acting	**Acting Humanly:** Turing Test approach (Alan Turing, 1950). Focuses on creating machines that perform actions requiring human intelligence when performed by people. Operates via text-based operational definitions.	**Acting Rationally:** Rational Agent approach. Focuses on designing agents that operate autonomously to achieve the best outcome (or maximum expected outcome under uncertainty). Standard is mathematically well-defined.

Turing Test vs. Total Turing Test

Standard Turing Test (Text-Only)

Interrogator queries a hidden human and machine via teletype. If the interrogator cannot distinguish machine from human, the machine passes.

- **Natural Language Processing (NLP):** Communicate in English.
- **Knowledge Representation:** Store knowledge/data.
- **Automated Reasoning:** Answer queries & infer rules.
- **Machine Learning:** Adapt to new circumstances.

Total Turing Test (Physical Interaction)

Includes visual signal & physical hatch for testing perceptual and physical manipulation capabilities.

- Includes all 4 standard capabilities.
- **Computer Vision:** Perceive objects & scenes.
- **Robotics:** Manipulate objects and move about in dynamic space.

2. TAXONOMY OF AI SYSTEMS

Type-1: Based on Capabilities

- **Weak / Narrow AI:** Trained exclusively for dedicated, specific tasks. Fails in unpredictable ways if pushed outside boundaries (e.g., Apple Siri, IBM Watson, Chess engines).
- **General AI:** Can perform ANY intellectual task with efficiency equal to a human. Smart & autonomous (Currently non-existent; under active research).
- **Super AI:** Hypothetical level where machines surpass human intelligence with cognitive reasoning, judgment, planning, and self-learning.

Type-2: Based on Functionality

- **Reactive Machines:** No memory/past experience stored. Reacts purely on current state (e.g., IBM Deep Blue, Google AlphaGo).
- **Limited Memory:** Stores short-term past data/experiences to inform immediate future actions (e.g., Self-driving cars tracking speed/distance of nearby vehicles).
- **Theory of Mind:** AI that understands human emotions, beliefs, and social interactions (Research phase).
- **Self-Awareness:** Sentient, conscious AI with internal emotions and self-awareness (Hypothetical future).

★ STRONG AI VS. WEAK AI VS. APPLIED AI VS. COGNITIVE AI

Strong AI: Self-aware machines capable of genuine cognitive mental states and reasoning indistinguishable from humans.
Weak AI: Simulates human cognition; acts as if intelligent without true mental states.

3. COMPLETE AI HISTORY & KEY MILESTONES

YEAR / PERIOD	MILESTONE EVENT	KEY PIONEERS & HISTORICAL SIGNIFICANCE
1943	First Artificial Neuron Model	Warren McCulloch & Walter Pitts (combined brain physiology, propositional logic, and Turing computation).
1949	Hebbian Learning Rule	Donald Hebb introduced neural connection strength updating rule.
1950	Turing Test & SNARC	Alan Turing published *"Computing Machinery and Intelligence"*; Minsky & Edmonds built SNARC (first neural net machine).
1955	Logic Theorist	Allen Newell & Herbert Simon created the first AI program (proved 38 of 52 math theorems).
1956	Dartmouth Conference	John McCarthy, Marvin Minsky, Claude Shannon & Nathaniel Rochester coined "Artificial Intelligence". AI born as a field.
1966	ELIZA Chatbot	Joseph Weizenbaum created the first NLP chatbot psychotherapist.
1972	WABOT-1 & Shakey	WABOT-1 built in Japan (first humanoid robot); Shakey (SRI) demonstrated mobile robot planning.
1974–1980	**First AI Winter**	Severe government funding cuts due to failure of machine translation, poor speech understanding, and neural net limits.
1980–1987	Expert Systems Boom	Emulated human expert decisions (e.g., XCON). AAAI held first national conference at Stanford (1980).
1987–1993	**Second AI Winter**	Collapse of specialized LISP hardware market; expert systems proved too costly to maintain.
1996–1997	EQP & Deep Blue	EQP proved Robbins Algebra is Boolean (1996); IBM Deep Blue defeated Garry Kasparov in Chess 3.5–2.5 (1997).
2002–2005	Roomba & STANLEY	iRobot Roomba entered homes (2002); Stanford's STANLEY won DARPA Grand Challenge driving 132 mi autonomously (2005).
2011–2016	Watson, Eugene & AlphaGo	IBM Watson won *Jeopardy!* (2011); Eugene Goostman passed Turing Test criteria (2014); AlphaGo beat Lee Sedol in Go (2016).

4. INTELLIGENT AGENTS: CORE CONCEPTS

Agent, Percept Sequence, Agent Function vs. Program

- **Agent:** Anything that perceives its environment through **sensors** and acts upon it through **actuators**. Formula: `Agent = Architecture + Program`.
- **Percept Sequence:** The complete history of everything the agent has ever perceived to date.
- **Agent Function (f):** Abstract mathematical mapping from percept sequence history to action ($f: P^* \rightarrow A$).
- **Agent Program:** Concrete software algorithm running on physical architecture that takes *only the current percept* as input.
- **Rational Agent:** Selects an action expected to maximize its performance measure based on the percept sequence and built-in knowledge. Rationality depends on: (1) Performance Measure, (2) Prior Knowledge, (3) Executable Actions, (4) Percept Sequence to date.

5. PEAS FRAMEWORK & SPECIFICATION EXAMPLES

AGENT TYPE	PERFORMANCE MEASURE	ENVIRONMENT	ACTUATORS	SENSORS
Automated Taxi Driver	Safe, fast, legal, comfortable trip, max profits, min wear/fuel	Roads, traffic, pedestrians, weather, customers, police	Steering, accelerator, brake, signals, horn, display	Cameras, sonar, speedometer, GPS, accelerometer, engine sensors
Medical Diagnosis System	Patient health, reduced treatment costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	Keyboard entry of symptoms, findings, patient answers
Part-Picking Robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and gripper hand	Camera, joint angle sensors
Satellite Image Analysis	Correct scene/image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays

6. PROPERTIES OF TASK ENVIRONMENTS

PROPERTY PAIR	DETAILED DEFINITION & OPERATIONAL MECHANICS	CONCRETE EXAMPLES
Fully vs. Partially Observable	Sensors access complete world state relevant to choice of action at all times vs. noisy, inaccurate, or missing data. Unobservable if no sensors exist.	Chess, Sudoku (Fully) vs. Automated Taxi, Vacuum without location sensor (Partially)
Single-Agent vs. Multi-Agent	Agent operates alone vs. operating with other entities that maximize performance measure based on Agent A's state (Competitive or Cooperative).	Crossword puzzle (Single) vs. Chess (Competitive Multi), Driving (Partially Cooperative Multi)
Deterministic vs. Stochastic	Next environment state is strictly determined by current state and action vs. outcomes having probabilistic uncertainty. (Nondeterministic = outcomes lack attached probabilities).	Vacuum world baseline (Deterministic) vs. Taxi driving, Vacuum with slipping suction (Stochastic)
Episodic vs. Sequential	Experience divided into atomic independent episodes (current action does not affect next episode) vs. short-term decisions having long-term consequences.	Defective assembly part picking (Episodic) vs. Chess, Taxi driving (Sequential)
Static vs. Dynamic vs. Semi-Dynamic	Environment constant while deliberating vs. continuously changing during decision-making vs. environment unchanged but performance score changes with time.	Crossword puzzles (Static) vs. Taxi driving (Dynamic) vs. Timed Chess (Semi-Dynamic)
Discrete vs. Continuous	Finite distinct states, time steps, percepts, and actions vs. continuously varying values over time.	Chess (Discrete) vs. Taxi speed, steering angles, continuous camera video (Continuous)
Known vs. Unknown	Agent or designer knows the "laws of physics" / rules of environment vs. agent must learn how environment operates.	Solitaire, Board Games (Known) vs. Video game on first playthrough (Unknown)

7. THE FOUR BASIC AGENT ARCHITECTURES

1. Simple Reflex Agent

Acts strictly on the **current percept**, ignoring percept history. Uses condition-action rules (`if condition then action`).

- **Requirement:** Requires fully observable environment to avoid infinite loops.
- **Escape Loops:** Randomizing actions allows escape from infinite state cycles.

2. Model-Based Reflex Agent

Handles partial observability by maintaining an internal **state** based on percept history.

- **Model Knowledge:** Encodes (1) How world evolves independently, (2) How agent's actions affect the world.

3. Goal-Based Agent

Combines state/model with explicit **goal information** describing desirable situations.

- **Advantage:** Highly flexible; behavior changes simply by updating goal state without rewriting rules.

4. Utility-Based Agent

Uses an internal **utility function** measuring degree of "happiness" or preference for states.

- **Advantage:** Allows optimal trade-offs under conflicting goals or uncertain outcomes (chooses max expected utility).

8. WELL-DEFINED SEARCH PROBLEM FORMULATION

★ 5 CORE COMPONENTS OF A SEARCH PROBLEM

1. **Initial State:** The starting state of the agent (e.g., `In(Arad)`).
2. **Actions:** Description of executable actions available to agent `ACTIONS(s)` .
3. **Transition Model:** Returns state resulting from action `RESULT(s, a)` .
4. **Goal Test:** Determines whether a given state is a goal state (explicit or implicit test).
5. **Path Cost:** Function `g(n)` assigning numeric cost to path; step cost denoted `c(s, a, s')` .

9. UNINFORMED (BLIND) SEARCH STRATEGIES

Uninformed search algorithms have no information regarding goal distance beyond the problem definition. Distinguished by node expansion order.

Breadth-First Search (BFS)
Expands shallowest unexpanded node first using a **FIFO Queue**. Goal test applied when node is **GENERATED**.

Uniform-Cost Search (UCS)
Expands node with lowest path cost `g(n)` using a **Priority Queue**. Goal test applied when node is **POPPED FOR EXPANSION**. Replaces frontier nodes if cheaper path found.

Depth-First Search (DFS) & Depth-Limited Search (DLS)
Expands deepest node using a **LIFO Queue / Stack**. Linear space complexity. DLS imposes limit `l` to avoid infinite paths.

Iterative Deepening Search (IDS)
Combines BFS completeness/optimality with DFS linear memory by gradually increasing depth limit (`l = 0, 1, 2, ...`). Preferred uninformed method for large search spaces with unknown solution depth.

10. MASTER UNINFORMED SEARCH COMPARISON TABLE

Let b = branching factor, d = depth of shallowest goal, m = max tree depth, l = depth limit, C* = optimal path cost, ε = min step cost.

ALGORITHM	FRONTIER DATA STRUCTURE	COMPLETE?	TIME COMPLEXITY	SPACE COMPLEXITY	OPTIMAL?
Breadth-First Search (BFS)	FIFO Queue	Yes (if b is finite)	$O(b^d)$	$O(b^d)$ (Memory bottleneck)	Yes (if step costs equal)
Uniform-Cost Search (UCS)	Priority Queue (ordered by g)	Yes (if step cost ≥ ε > 0)	$O(b^{1 + \lfloor C^*/\varepsilon \rfloor})$	$O(b^{1 + \lfloor C^*/\varepsilon \rfloor})$	Yes (general path cost)
Depth-First Search (DFS)	LIFO Queue / Stack	No (fails in infinite depth)	$O(b^m)$	$O(b \cdot m)$ (Linear space)	No
Depth-Limited Search (DLS)	LIFO Stack with limit l	No (if l < d)	$O(b^l)$	$O(b \cdot l)$	No
Iterative Deepening (IDS)	LIFO Stack (repeated DLS)	Yes (if b is finite)	$O(b^d)$	$O(b \cdot d)$ (Optimal memory)	Yes (if step costs equal)
Bidirectional Search	2× Frontiers (Forward & Backward)	Yes (if both use BFS)	$O(b^{d/2})$	$O(b^{d/2})$	Yes (if unit step costs)

11. EXAM NIGHT REVISION CHECKLIST & HIGH-YIELD DISTINCTIONS

★ **CRITICAL EDGE CASES & FREQUENTLY TESTED EXAM QUESTIONS**

- **Goal Test Timing Difference:**
  - **BFS:** Applies Goal Test upon *GENERATION* of child node (saves expanding unnecessary level).
  - **UCS:** Applies Goal Test upon *SELECTION / POPPING FOR EXPANSION* (guarantees optimal path cost).
- **Why IDS is preferred over BFS:** Re-generating upper nodes in IDS adds only a small overhead (b / (b - 1) ratio), but reduces space from exponential $O(b^d)$ down to linear $O(b \cdot d)$.
- **Tree Search vs Graph Search in DFS:** Graph Search avoids infinite loops in finite spaces by keeping an explored set; Tree Search fails on infinite non-goal paths.
- **Deterministic vs Stochastic vs Nondeterministic:** Stochastic uses quantified probabilities; Nondeterministic has unquantified outcomes and requires success across all outcomes.
- **Semi-dynamic vs Dynamic Environment:** Dynamic changes continuously (Taxi driving); Semi-dynamic means world state doesn't change, but agent score changes with time (Timed Chess).